Next Bill Amount Prediction

Approach and defence guide: what I did, how, why, and why not the alternatives

Use case: predict every customer's next bill one billing cycle ahead, explain it, and flag unusually high bills.

Data: 2000 billing records, 1222 customers, 35 columns, Jan 2024 to Aug 2026 (synthetic).

Companion files: Next_Bill_Amount_Prediction.ipynb (all code and outputs) and the 6-slide presentation.

The project in one page

Problem. A billing company wants to know each customer's next bill early, so that sudden increases (bill shock) can be reviewed before the customer calls. The model must predict a number (regression), use only information available at prediction time, be tested on a later period, explain itself, and flag unusually high predicted bills.

Approach in six lines

- **Split by time:** train on Jan 2024 to Jan 2026 (1563 records), test on Feb to Aug 2026 (437 records).
- **Clean on training data only:** clip extreme values, one-hot encode, remove duplicated information, scale inside pipelines.
- **Beat simple rules first:** five baselines (previous bill, recent averages, plan median, mean).
- **Compare ten models** with time-based cross-validation; tune the four most promising.
- **Judge on MAE** (typical error), with RMSE, R^2 and WAPE next to it; test months touched only once.
- **Explain and monitor:** permutation importance, customer-level drivers, a high-bill flag and 80% prediction intervals.

Headline result (held-out Feb to Aug 2026)

	MAE	RMSE	R^2	WAPE
Tuned SVR (final model)	**163.5**	328.7	0.508	20.2%
Plan-type median (best baseline)	179.1	335.0	0.489	22.1%
Average of last 6 bills	184.7	332.8	0.496	22.8%
Previous bill only	225.2	401.9	0.265	27.8%

- About **9% lower MAE** than the best baseline, and the bootstrap interval for the gain (6.7 to 24.1) excludes zero, so it is reliable. It is better than the baseline in all 7 test months.
- The RMSE gain is **not** statistically clear (interval -7.1 to 18.9): both model and baseline miss the big spikes.
- The bills are driven mostly by the plan, recurring charges and recent billing history; last cycle's usage adds almost nothing.
- The high-bill flag is only **weakly** better than random (ROC-AUC 0.565). I say this openly: it is a review list, not an alarm.

Step-by-step reasoning

Every step has the same four questions. **WHAT** is done, **HOW** it is done, **WHY** this choice, and **WHY NOT** the obvious alternatives. This is the part to rehearse for the viva.

1 Understand the problem and check the data for leakage

WHAT	Define the target (*next_bill_amount*, the bill after the latest one) and the forecast horizon (one bill ahead).
HOW	Grouped the input columns into plan, usage, latest-bill charges and billing history. Checked that the latest bill equals its components plus tax (largest difference 0.0) and that the 3-month average is built from earlier bills.
WHY	A model is only valid if every input is known at prediction time. The check proves no input contains the answer, so there is **no target leakage**.
WHY NOT	Not "feed all columns and see what happens": one column that secretly contains the target would give a fantastic score that is useless in real life.
RESULT	2000 records, 1222 customers (1 to 3 records each), Jan 2024 to Aug 2026.

2 Data quality checks

WHAT	Look for duplicates, missing values, wrong types and impossible values.
HOW	Shape, *info()*, *describe()*, duplicate count, missing-value count.
WHY	Dirty data silently damages every later step, so it is checked first.
WHY NOT	No imputation and no row deletion: there were 0 duplicates and 0 missing values, so nothing needed fixing. Inventing fixes would only add noise.
RESULT	The data is clean.

3 Exploratory data analysis (EDA)

WHAT	Understand the target, the features, the relationships between them and the behaviour over time.
HOW	Histograms, boxplots and a skewness table; correlation heatmaps; scatter plots against the target; boxplots by segment, region and plan; monthly trends; one customer's bill history; two extra checks (seasonality, how often a bill repeats an old bill).
WHY	EDA decides the modelling choices. The bill is right-skewed with spikes, so a robust loss and MAE make sense; the bill follows the plan fee and the history, so plan type and history features must be kept.
WHY NOT	EDA is used only to *describe*. No cleaning limit or feature decision is taken from it, because anything learned from the test months would leak.
RESULT	The next bill equals one of the last 3 bills in **45.1%** of the records (very stable customers). Next bill / (plan price x 1.18) has median 1.15 and 25th percentile 1.00: most bills sit at or above plan price plus tax, with a long tail up to 9.9. No clear seasonality: calendar-month averages go from about 755 (Feb) to 933 (Oct), but this comes from a few very large bills.

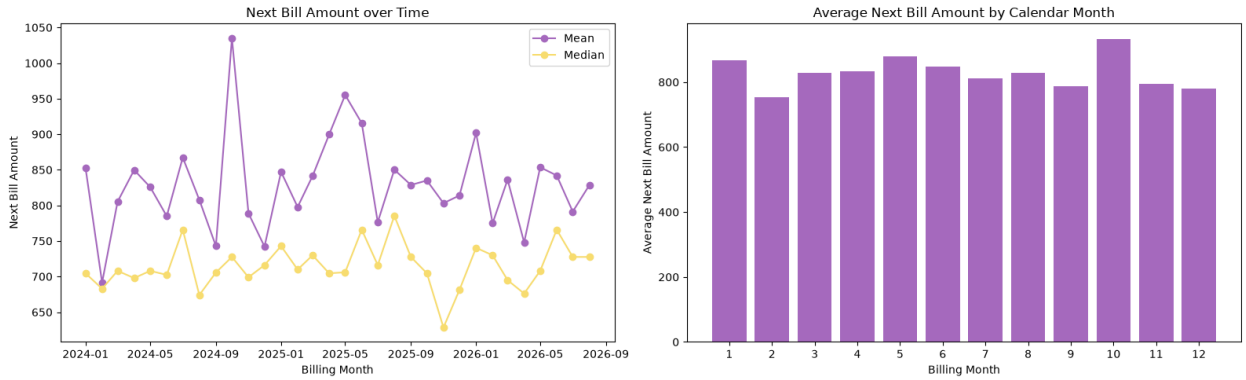


Figure 1. Average next bill over time and by calendar month: noisy, no clear seasonal pattern.

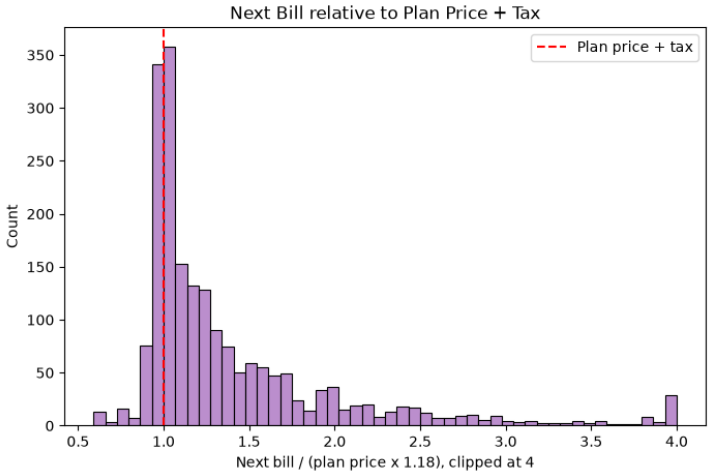


Figure 2. Next bill relative to plan price plus tax: a floor near 1.0 and a long right tail of spikes.

4 Chronological train-test split

WHAT	Train on the first 25 months, test on the last 7 months.
HOW	Sort by billing month, take the first 80% of months for training (Jan 2024 to Jan 2026, 1563 records) and the last 20% for testing (Feb to Aug 2026, 437 records). An assertion confirms the last training month is earlier than the first test month.
WHY	In real life the model predicts the future from the past. The test must copy that situation, otherwise the score does not mean anything.
WHY NOT	**Not a random split** (also not a random split by customer): it puts future months into training and past months into the test set, so the model "already knows" the future and the scores become optimistic. My first version of the notebook used a random grouped split; I replaced it for this reason.
RESULT	54% of test records belong to customers that also appear in training, which is realistic for a billing company.

5 Preprocessing, fitted on the training months only

WHAT	Outlier clipping, encoding, removal of redundant features and scaling.

HOW	**Outliers:** 26 features are clipped at the 1st and 99th percentile of the training data; zero-heavy features (roaming, overage, late fee) keep their zeros; the target is never clipped. **Encoding:** one-hot encoding (first category dropped) for segment, region and plan type, fitted on training data. **Redundancy:** removed *overage_data_gb* (correlation 0.9997 with the data overage charge), *plan_price* (plan type already gives one price per plan; 0.98 correlated with the previous plan price) and *billing_year*. **Scaling:** *StandardScaler* inside the pipelines of Ridge, Lasso and SVR, so it is refitted on every training fold.
WHY	Anything learned from data (clip limits, categories, scaling means) must come from the training months, otherwise test information leaks into training. Clipping stops a few extreme values from dominating scale-sensitive models. *billing_year* is only a time counter and 2026 is mostly outside the training years.
WHY NOT	**Not deleting outlier rows:** only 1563 training rows, and the spikes are real bills the system must learn to handle; an IQR rule would delete 23% of rows for one feature. **Not clipping the target:** we must predict real bills. **Not label encoding:** it invents an order between regions and plans. **Not scaling on all data:** that is leakage. Trees do not need scaling at all.
RESULT	The final feature matrix has 45 columns, and 0 missing values.

6 Feature engineering, feature selection and PCA

WHAT	Test whether new, selected or compressed features improve the model.
HOW	**Engineering:** 9 new features from known past values (bill volatility, bill relative to base charge, recurring vs variable charges, usage trend, a "flat history" flag, month sine and cosine). **Selection:** keep the top half of features by Random Forest importance, selected inside every cross-validation fold; Lasso and Random Forest importance were also plotted. **PCA:** keep 95% of the variance. A rule decided in advance: adopt a change only if cross-validated MAE improves by more than 1% and RMSE does not get worse.
WHY	A feature has to earn its place. More features mean more noise and a model that is harder to explain.
WHY NOT	Not keeping the new features "because they sound clever": the evidence said no.
RESULT	New features: MAE improvement **0.03%**, RMSE 0.28%. Not adopted. Selection: MAE improvement **0.14%**, RMSE 0.41%. Not adopted, all 45 features kept. PCA needs 27 components for 95% variance and Ridge gets worse (CV MAE 216.9 vs 213.8). Not used.

7 Validation strategy and evaluation metrics

WHAT	Choose how models are compared and which numbers decide.
HOW	Time-based cross-validation inside the training months: **5 expanding-window folds**. Each fold trains on all earlier months and validates on the next 3 to 4 months (the first fold has 557 training rows; the last validates on Nov 2025 to Jan 2026). Metrics: MAE, RMSE, R^2, adjusted R^2, WAPE.
WHY	Models can be compared fairly without ever touching the test months, and the time order is respected inside the training data too. **MAE is the main metric** because bills have spikes (skewness about 2) and MAE is the typical error in money. WAPE (total absolute error / total bill) gives a percentage that is easy to explain. RMSE is reported because it punishes big misses.

WHY NOT	**Not shuffled K-fold:** it trains on the future to predict the past. **Not choosing models on the test set:** then the test is no longer unseen. **Not R^2 alone:** a few huge bills dominate it.

8 Baselines before any machine learning

WHAT	Five simple rules: training mean, plan-type median, previous bill, average of last 3 bills, average of last 6 bills.
HOW	Evaluated on exactly the same validation folds and the same test months (using the original, unclipped billing columns).
WHY	A model is only useful if it beats what a clerk could do with a spreadsheet. Without baselines "R^2 = 0.5" means nothing.
WHY NOT	Not skipping them, and not picking the best baseline by looking at the test set: it was chosen by cross-validation.
RESULT	Best baseline: **plan-type median** (CV MAE 206.6, test MAE 179.1, R^2 0.489). Average of last 6 bills is close (test MAE 184.7). The previous bill alone is poor (R^2 0.265) because single bills contain spikes.

9 Which models, and why these

WHAT	Ten regression models: Linear Regression, Ridge, Lasso, Decision Tree, Random Forest, Gradient Boosting, Gradient Boosting (Huber loss), XGBoost, XGBoost (MAE objective) and SVR.
HOW	All trained with default-style settings and compared with time-based cross-validation, then once on the test months.
WHY	Tabular regression with about 1500 rows: linear models give an explainable benchmark, trees and boosting capture non-linear effects, SVR is a smooth kernel model. Because bills contain spikes, two **robust-loss versions** were added (Huber loss and absolute-error objective); they care about the typical bill, not the extremes.
WHY NOT	**ARIMA, Prophet, LSTM:** customers have only 1 to 3 records, so there is no real time series per customer; deep learning also needs far more data and is hard to explain. **Classification models** (e.g. logistic regression): the target is a number. **CatBoost and LightGBM:** very similar to XGBoost; no extra library was needed. **Log-transformed target:** in an early check outside the notebook it lowered MAE for some models but made SVR much worse (CV MAE about 270 to 300 against 197), so I used robust losses instead. This is a possible extension.

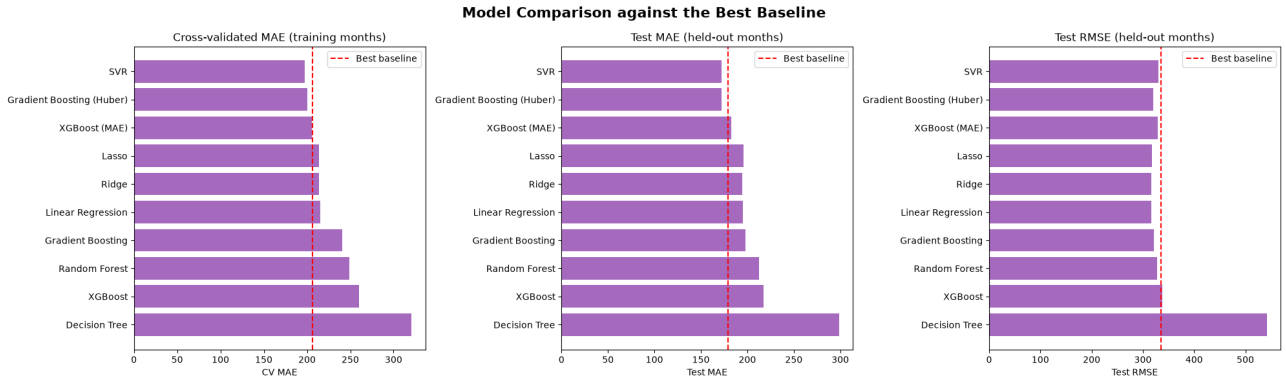


Figure 3. Cross-validated MAE, test MAE and test RMSE of all models. The red line is the best baseline.

Model	CV MAE	Train MAE	Test MAE	Test RMSE	Test R^2
SVR	197.5	121.7	172.3	329.8	0.505
Gradient Boosting (Huber)	200.3	146.9	172.3	320.7	0.532
XGBoost (MAE)	205.7	76.4	182.8	329.0	0.507
Lasso	213.5	199.1	195.6	317.6	0.541
Ridge	213.8	198.1	194.3	315.8	0.546
Linear Regression	215.5	198.0	195.1	317.0	0.542
Gradient Boosting	240.3	157.7	198.2	321.7	0.529
Random Forest	248.7	85.8	212.7	327.6	0.511
XGBoost	260.4	20.1	217.5	338.2	0.479
Decision Tree	320.7	0.0	298.7	541.6	-0.335

Table 1. Default-setting models, ordered by cross-validated MAE. Models with a robust loss win on MAE; the linear models win on RMSE and R^2. Untuned Decision Tree, XGBoost and Random Forest overfit (training MAE far below validation MAE).

10 Hyperparameter tuning

WHAT Tune the four most promising models by cross-validated MAE: SVR, Gradient Boosting (Huber), XGBoost (MAE) and Lasso.

HOW *GridSearchCV* for small grids (SVR, Lasso) and *RandomizedSearchCV* with 30 draws for the boosting models; the same time-based folds and MAE as scoring.

WHY Default settings are rarely best (untuned XGBoost has a training MAE of 20 but a validation MAE of 260). Using the same folds and metric means the search cannot see the test months.

WHY NOT **Not tuning all ten models:** little value for plain linear regression or a single tree.
Not tuning on the test set: that would leak.
Not Optuna or huge grids: extra library and more risk of fitting the folds; random search is enough for small data and easier to explain.

RESULT Tuning lowered the cross-validated MAE of all four, and the test MAE moved the same way (table below). Tuned CV scores are slightly optimistic because the best setting was picked on the same folds.

Model	CV MAE default	CV MAE tuned	Test MAE default	Test MAE tuned	Test R^2 tuned
SVR	197.5	184.5	172.3	163.5	0.508
XGBoost (MAE)	205.7	187.0	182.8	167.6	0.519
Gradient Boosting (Huber)	200.3	190.0	172.3	168.8	0.517
Lasso	213.5	212.5	195.6	194.0	0.544

Table 2. Default versus tuned models.

11 Final model selection: tuned SVR

WHAT Choose one model for the final evaluation: SVR with C = 1000, epsilon = 10, gamma = 0.003.

HOW A rule fixed in advance: the tuned model with the lowest cross-validated MAE (SVR 184.5, XGBoost (MAE) 187.0, Gradient Boosting (Huber) 190.0). Then evaluated once on the test months and compared with the best baseline by a paired bootstrap (2000 resamples).

WHY	SVR minimises an absolute-type loss (errors below epsilon are ignored, large errors count linearly), which is exactly right for bills with spikes. It had the lowest cross-validated and test MAE, shows no overfitting (train MAE 157.0, test MAE 163.5) and beats the baseline in all 7 test months.
WHY NOT	**Not Ridge or Lasso:** best RMSE and R^2 (about 0.54) but a clearly higher MAE (about 194). **Not XGBoost or Gradient Boosting:** within about 5 MAE of SVR in cross-validation, so it is a close call; SVR was slightly better on both validation and test and I fixed the rule before looking at the test months.
RESULT	Test MAE **163.5** vs 179.1 baseline, WAPE 20.2% vs 22.1%, R^2 0.508 vs 0.489. MAE reduction 15.5 with 95% interval [6.7, 24.1] (reliable); RMSE reduction 6.4 with interval [-7.1, 18.9] (not clear). **Honest trade-off:** SVR is best on MAE and WAPE, the linear models are best on RMSE and R^2.

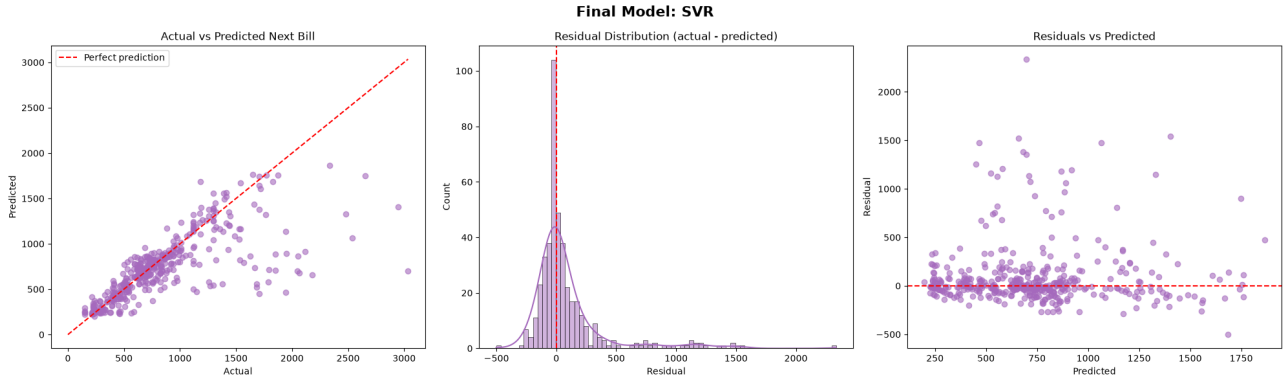


Figure 4. Final SVR: actual vs predicted, residual distribution and residuals vs predicted. Most predictions are close; a group of bills is far above the prediction.

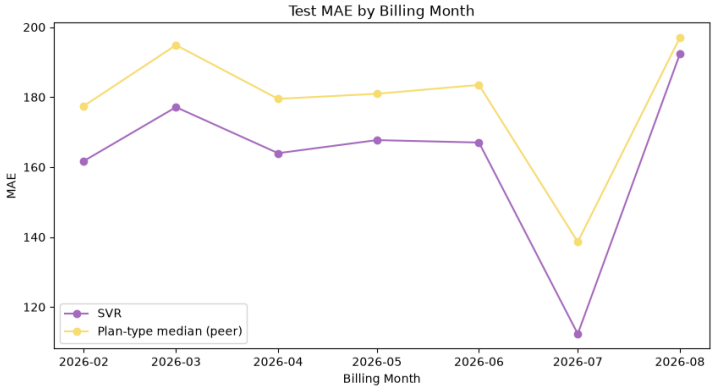


Figure 5. Test MAE by month: the model is below the baseline in all seven months (smallest gap in August).

12 Segmented performance and error analysis

WHAT	Find where the model works well and where it fails.
HOW	MAE and WAPE by previous-bill range, actual bill range, customer tenure, plan type, segment and region, compared with the best baseline; the ten largest errors; the share of total error from the worst records.
WHY	One overall number hides weak groups, and the brief asks for segment-wise reliability.
WHY NOT	Not trusting small segments: Family 749 (19 test records), Unlimited 1399 (8) and North-East (13) look worse than the baseline, but the samples are too small to conclude.

RESULT	Better than the baseline in every previous-bill range (3% to 15%); biggest gains for tenure 8+ years (19%) and Value 349, Lite 199, Unlimited 649 plans. **Weak point:** bills above 1500 (MAE about 817, WAPE 42.8%, no better than the baseline). The ten largest errors are all **under-predicted spikes** (actual 1.6 to 4.5 times the 6-month average); the worst 10% of records give **53.6%** of all error. Nothing in the previous-cycle data warns about these jumps; this is the limit of the data, not a coding mistake.

13 Interpretation: what drives the bill

WHAT	Explain the model globally (portfolio) and for each customer.
HOW	**Global:** permutation importance on the test months (how much MAE worsens when one feature is shuffled), also summed by feature group. **Per customer:** each feature is set to its typical (training median) value and the change in the prediction is measured; the three biggest changes are shown as the customer's drivers.
WHY	The brief needs both portfolio-level and customer-level drivers. Both methods work for any model and are simple to explain.
WHY NOT	**Not model coefficients:** an RBF-kernel SVR has none. **Not SHAP:** an extra library and slow for kernel models; the median-replacement method answers the same question simply. Explanations show **influence on the prediction, not proven billing causality**; correlated features share importance (for example tax amount and previous bill).
RESULT	Top features: *base_charge* (+36.7 MAE when shuffled), *average_bill_last_6_months* (+35.9), plan type Lite 199 (+26.3). By group: billing history, plan type and recurring plan charges explain almost everything; usage, overage, roaming, late fees and calendar month add almost nothing.

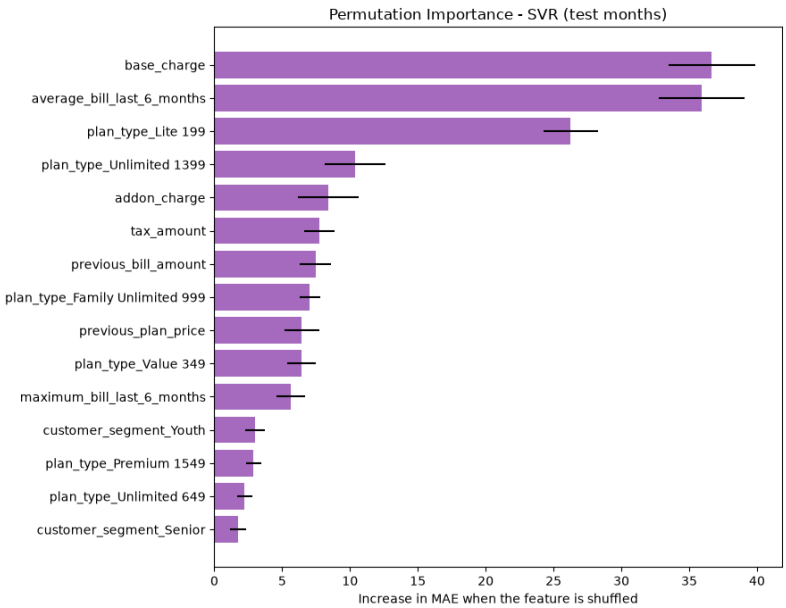


Figure 6. Permutation importance on the test months (top 15).

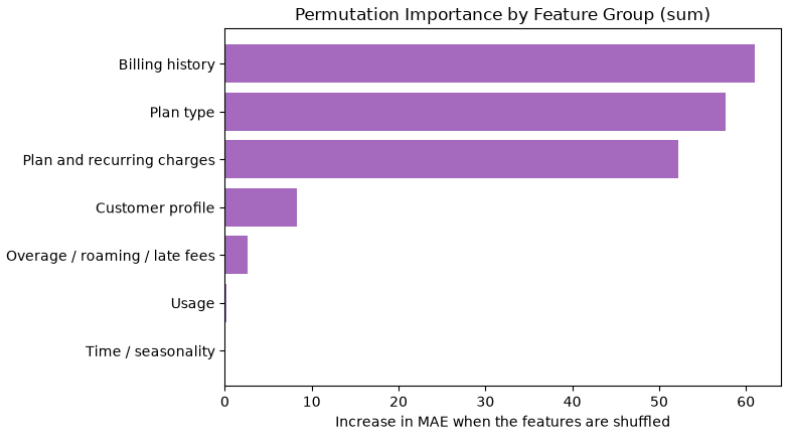


Figure 7. Importance by feature group.

14 High-bill flag and prediction intervals

WHAT	Flag customers whose predicted bill is unusually high, and attach an uncertainty range.
HOW	**Definition:** a high bill is an actual bill at least 1.5 times the customer's own 6-month average (11.2% of test records, 49 of 437). **Flag rule:** predicted bill >= t x 6-month average. The multiplier t = 1.1 was chosen on out-of-fold training predictions (best F1), never on test. **Uncertainty:** two Gradient Boosting models for the 10th and 90th percentile give an 80% interval. **Monitoring view:** flagged customers ranked by predicted rise over their own average, with prediction, interval, history and top drivers.
WHY	The brief asks for a documented rule, precision and recall against a documented definition, and an uncertainty measure. A personal baseline (own 6-month average) is fairer than a global limit.
WHY NOT	**Not a fixed global limit:** a 1500 bill is normal for a premium plan but shocking for a 199 plan. **Not choosing t on the test set:** leakage. Not claiming it works better than it does (below).
RESULT	Flag at t = 1.1: 51 customers flagged, precision **0.18**, recall **0.18**. Without a threshold: ROC-AUC **0.565** (naive previous-bill score 0.540, random 0.5); precision among the top 25 is 20% against a base rate of 11.2%. Interval coverage 75.5% (target 80%), average width about 507: honest but wide. The interval is slightly too narrow because of the upward spikes. **Conclusion:** use the flag to prioritise reviews, not as an alarm.

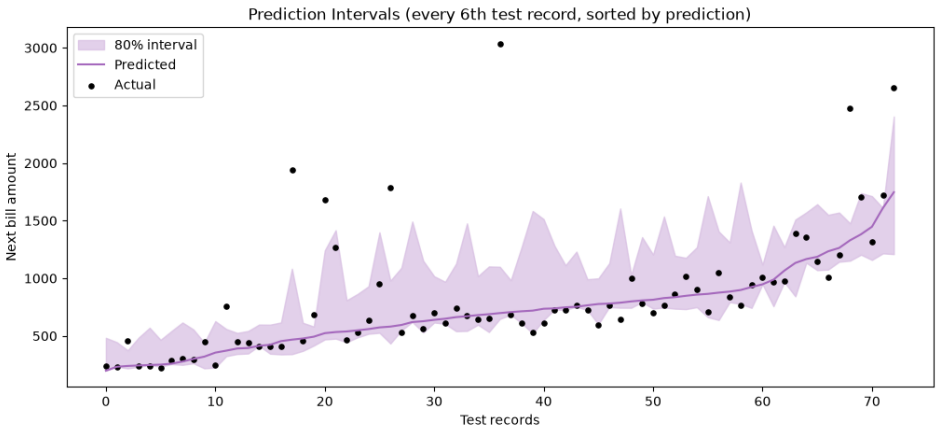


Figure 8. 80% prediction intervals (every 6th test record, sorted by prediction). Black dots above the band are unforeseen spikes.

15 Limitations and next steps

WHAT	State clearly what the project cannot claim.
HOW	Listed in the notebook conclusion.
WHY	Honest limits make the result more credible in a viva or interview.
WHY NOT	Not hiding the weak flag or the unclear RMSE gain.
RESULT	Only 2000 records and 1 to 3 per customer: no real customer time series; some segments have very few test records. Synthetic data: the spikes may be random by construction. Real billing data must be re-validated. Tuned CV scores are slightly optimistic; clip limits use the whole training period. **Next steps:** longer monthly history per customer, a separate classifier for bill-shock probability, a log-target model, and periodic retraining on a rolling window.

Likely questions and short answers

Q. Why a chronological split and not a random one?

A bill forecast must predict the future from the past. A random split lets the model train on months after some test records, which makes the score look better than it would be in real use.

Q. Why did you choose MAE as the main metric?

Bills have random spikes (skewness about 2). MAE is the typical error in money and is not dominated by a few extreme bills. RMSE, R^2 and WAPE are reported next to it.

Q. Why SVR when Ridge has a better R^2?

The rule was fixed beforehand: lowest cross-validated MAE. SVR has the lowest MAE and WAPE; Ridge reacts more to large bills so it has better RMSE and R^2. I state this trade-off openly.

Q. Is the model overfitted?

No. Train MAE is 157.0 and test MAE is 163.5, and it beats the baseline in all seven test months. The untuned Decision Tree, XGBoost and Random Forest are the overfitted ones, which is why tuning and time-based validation were used.

Q. Is 8.7% better than the baseline worth it?

The MAE gain is statistically reliable (bootstrap interval 6.7 to 24.1), but modest. The data contains spikes that nothing in the inputs predicts, which limits every model. I do not claim more than that.

Q. How do you know there is no data leakage?

Every input is a past or current-cycle value (the previous bill equals its components plus tax), the split is chronological, and all clip limits, encoders, scalers, thresholds and selections are fitted on training data or inside the training folds.

Q. Why not delete outliers?

Spikes are real bills. With only 1563 training rows, deleting them would remove exactly what the business wants to anticipate. They are clipped at the 1st and 99th percentile instead, and the target is untouched.

Q. Why did feature engineering not help?

The nine new features changed cross-validated MAE by 0.03%. The original features (history, base charge, plan type) already hold that information. I kept the simpler model.

Q. Why not LSTM, ARIMA or Prophet?

Each customer has only 1 to 3 records, so there is no real time series per customer. Deep learning needs far more data and is hard to explain.

Q. What is WAPE?

Weighted absolute percentage error: total absolute error divided by total actual bill. Final model: 20.2%, best baseline: 22.1%.

Q. How reliable is the high-bill flag?

Only weakly. ROC-AUC 0.565 and precision 0.18 at recall 0.18. It is about 1.4 to 1.8 times better than random, so it works as a prioritised review list, not as an alarm.

Q. Do your explanations prove what causes a high bill?

No. They show which inputs influence the model's prediction. Correlated features share importance, so this is influence, not causality.

Q. How would you improve it?

More monthly history per customer, a separate bill-shock classifier, a log-target model, and retraining on a rolling window.

Glossary

Term	Meaning
MAE	Mean absolute error: the average size of the prediction error, in the same units as the bill.
RMSE	Root mean squared error: like MAE but large errors count much more.
R^2	Share of the variation in bills explained by the model (1 is perfect, 0 is no better than the mean).
WAPE	Total absolute error divided by total actual bill, as a percentage.
Baseline	A simple rule (e.g. "next bill = previous bill") the model has to beat.
Cross-validation	Repeatedly training on part of the training data and validating on the rest to compare models fairly.
Time-based (expanding window) CV	Cross-validation where each fold trains only on earlier months and validates on the following months.
Data leakage	Information from outside the training period or from the target slipping into the model, which makes scores unrealistically good.
Winsorizing / clipping	Replacing values beyond chosen percentiles with those percentile values instead of deleting rows.
Robust loss (Huber / absolute error)	A loss that gives large errors less weight than squared error, so spikes disturb the model less.
Permutation importance	How much the error increases when one feature is randomly shuffled.
Prediction interval	A range expected to contain the actual bill with a stated probability (here 80%).